

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

DEPRESSION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACH Anticholinergic

ADR Adverse Drug Reaction

βB Beta Blockers

CBC Complete Blood Count

CV Cardiovascular

DAA Dual Action Antidepressants

DSM Diagnostic & Statistical Manual

ECT Electroconvulsive Therapy

GI Gastrointestinal

IPT Interpersonal Therapy

MAOI Monoamine Oxidase Inhibitor

MOA Mechanism Of Action

mCBT Mindfulness-based Cognitive Behavioural Therapy

MDD Major Depressive Disorder

NDRI Norepinephrine Dopamine Reuptake Inhibitor

NSAID Non-Steroidal Anti-Inflammatory Drug

N/V Nausea/Vomiting

PDE5 Phosphodiesterase 5

PHQ Patient Health Questionnaire

PPD Post-Partum Depression

PPI Proton Pump Inhibitor

QIDS-SR Quick Inventory of Depressive Symptomatology

rTMS Repetitive Transcranial Magnetic Stimulation

SERT Serotonin Transporter

SIADH Syndrome Of Inappropriate Antidiuretic Hormone Secretion

SNRI Serotonin Norepinephrine Reuptake Inhibitor

SSRI Selective Serotonin Reuptake Inhibitor

TCA Tricyclic Antidepressants

DEFINITION

Depressive Disorders include:

- Persistent Depressive Disorder (Dysthymia)
- Major Depressive Disorder
- Depressive Disorder not otherwise specified

Depression A state of low mood that can affect emotional, physical, cognitive and behavioural well-being

Persistent Depressive Disorder

For **≥ 2 years**, a person has experienced a ***depressed mood*** more days than not, **PLUS ≥ 2** of the following:

- Appetite loss or overeating
- Sleep disturbances
- Low energy/fatigue
- Low self-esteem
- Hopelessness
- Poor concentration or indecisiveness

Major Depressive Disorder

≥ 5 of the following symptoms for at least **2 weeks**; at least **1** must be ***depressed mood or loss of interest/pleasure***:

- Depressed mood
- Markedly reduced interest or pleasure in all activities
- Appetite loss or overeating
- Sleep disturbances
- Agitation or psychomotor retardation
- Fatigue/low energy
- Feeling worthless or excessive/inappropriate guilt
- Poor concentration
- Recurrent thoughts of death

MEMORY AID: MDD CLINICAL PRESENTATION

S	Sleep changes
A	Anhedonia
D	Depressed mood
A	Appetite disturbance
F	Fatigue
A	Agitation (psychomotor) or psychomotor retardation
C	Concentration (diminished ability)
E	Esteem (low)
S	Suicidal ideation

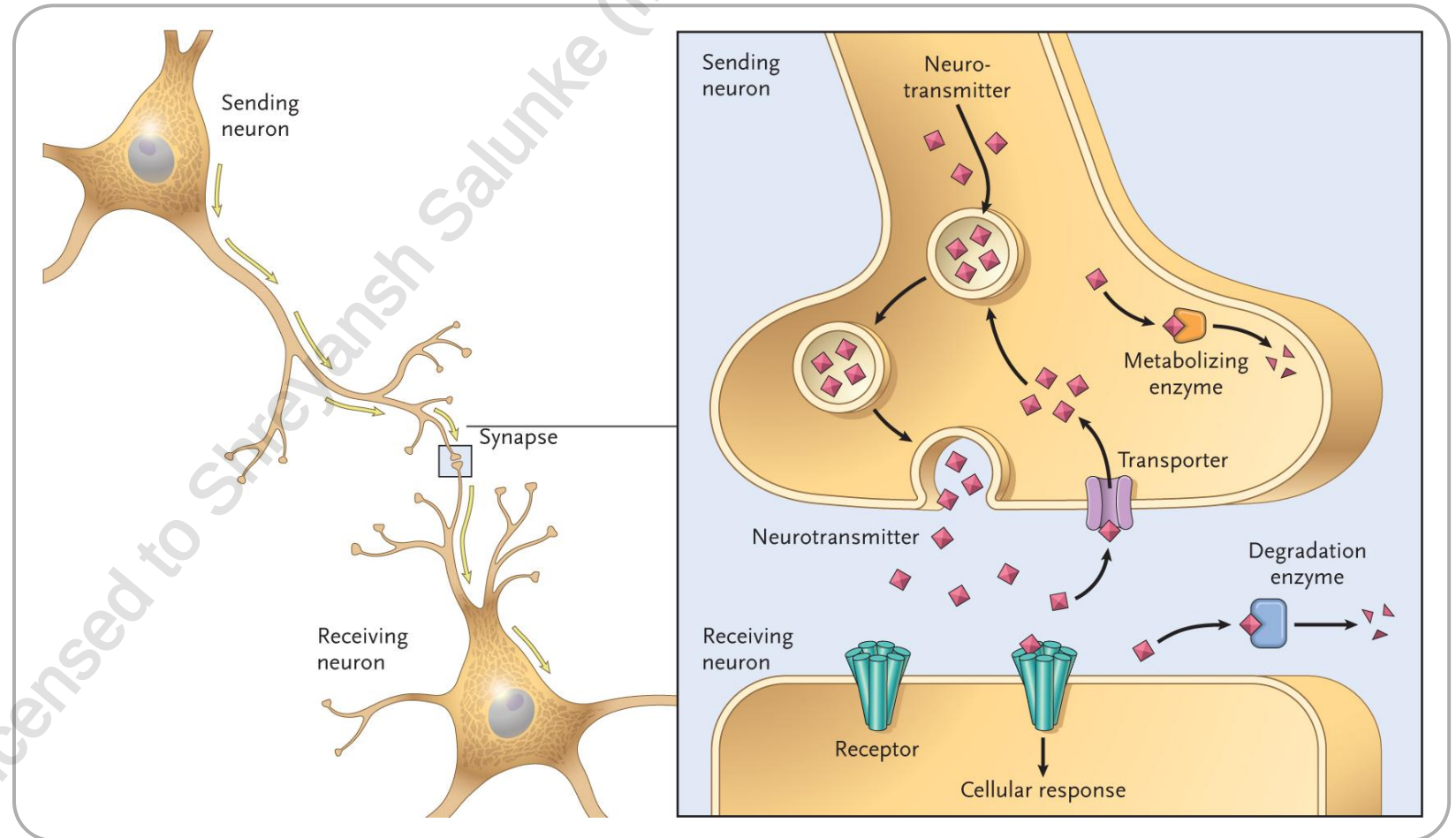
"SADAFACES"



PATHOPHYSIOLOGY

Postulated Monoamine Theory of Depression

- Depression may be caused by **underactivity** at **noradrenergic** and/or **serotonergic synapses**
- However:
 - **Anti-depressants do not cause a response immediately**, it takes at least 2-4 weeks for antidepressants to show any therapeutic effect & can take **6-8 weeks to show full therapeutic effect**



EXAMPLES OF SCREENING TOOLS AVAILABLE FOR SCREENING DEPRESSION

**Patient Health
Questionnaire
(PHQ-9)**

Self-administered tool

**Hamilton
Depression Rating
Scale (HAM-D-7)**

Physician assessment tool measuring
severity of depression

**Edinburgh
Postnatal
Depression Scale
(EPDS)**

Used in pregnant females and in post-
partum depression

RISK FACTORS

Depression

Nonmodifiable

- Females (males catch up later in life)
- Family history (first degree relative)
- History of adverse childhood experiences, traumatic life events, death of spouse

Potentially modifiable

- Chronic medical conditions and psychiatric comorbidities e.g. diabetes, cardiovascular disease, anxiety, chronic pain
- Alcohol and substance use disorders
- Periods of hormonal changes
- Significant life stressors
- Bereavement
- Environmental/lifestyle factors

Suicide



Comorbid conditions

- Posttraumatic stress disorder (PTSD)
- Substance use disorder (or family history)
- Comorbid personality disorders
- Sleep disorders, chronic pain conditions

Life events

- Suicidal ideation, prior suicide attempt
- Hopelessness, anxiety, impulsivity, psychosis
- Significant life stressors
- Increasing age, being widowed or unmarried
- Unemployed or living alone
- Anniversary of a loss
- Ethnicity → e.g. First Nations
- Females are more prone to overdose

SUICIDE SAFETY PLAN

A 6-step collaborative plan should be made in individuals deemed at high risk of dying by suicide to help patients self-manage vulnerable episodes they may come across in the future

Step 1	Identify Warning Signs	Thoughts, moods, and circumstances that may lead to suicidal ideation
Step 2	Enlist Personalized Coping Strategies	Self-management strategies that patient and care team decide on (meditation, working out, journaling, etc.)
Step 3	Enable Distraction and Connect with Others	Do something with a friend, go out to eat, visit online forums for support, etc.
Step 4	Engage with Social and Community Supports	Personal contacts should be involved in contingency plans
Step 5	Identify Professional Contacts	Mental health provider, crisis help lines, urgent care, emergency room, etc.
Step 6	Make the Environment Safe	Remove excess medications, firearms and other weapons, avoid busy streets and heights

DRUG-INDUCED CAUSES

- Anticholinergic drugs
- Anticonvulsants
 - Levetiracetam, phenobarbital, primidone, phenytoin, tiagabine, topiramate, vigabatrin
- **Antipsychotics**
- Antivirals
- Beta blockers (especially lipophilic molecules)
- Calcium channel blockers (nifedipine, diltiazem and verapamil)
- Clonidine
- Corticosteroids
- **Drugs for Parkinson's Disease**
- Interferon alpha/beta
- **Isotretinoin**
- Methyldopa
- Proton Pump Inhibitors (PPIs)
- Histamine 2 Receptor Antagonists (H2RAs)
- Reserpine
- **Sex hormones** (e.g. oral contraceptives, tamoxifen, GnRH agonists)
- Statins
- Stimulants
- Triptans
- Varenicline

If a patient has a history of depression, are drugs that cause depression contraindicated?

How do we handle situations where a patient with depression is prescribed a drug that may cause depression?

GOALS OF THERAPY

- ✓ Attain relief from depressive symptoms
- ✓ Treat concurrent symptoms/disorders with minimal side effects
- ✓ Restore optimal functioning and quality of life
- ✓ Prevent suicide
- ✓ Prevent future episodes

NON-PHARMACOLOGICAL MEASURES

Psychoeducation and self-management (CBT, IPT, and BA have most evidence)

- Preferred over pharmacotherapy for **mild depression** as it is as effective
- In **moderate depression**, may start with psychotherapy **OR** pharmacotherapy
- In **severe depression** patients should receive BOTH psychotherapy **AND** pharmacotherapy
- Psychotherapy combined with pharmacotherapy is **more effective** than either alone
- Educate on symptoms, coping strategies, and treatment options to facilitate shared decision-making
- Empower patient by teaching methods to actively manage their depression and maintain a good quality of life (e.g. problem-solving, lifestyle modification, and behavioral activation)

Regular Exercise (Aerobic and resistance)

- Physical activity is proven to prevent and treat depression
 - Supervised exercise, low intensity for 30-40 min at a time, 3-4x weekly for a minimum of 9 weeks
- First-line monotherapy option for **mild depression**
- May be used as a second-line adjunct in **moderate depression**
- Promote enjoyable physical activity experiences and slowly build up activity
 - Play outdoors, incorporate music, participate with a friend or group, etc.

NON-PHARMACOLOGICAL MEASURES

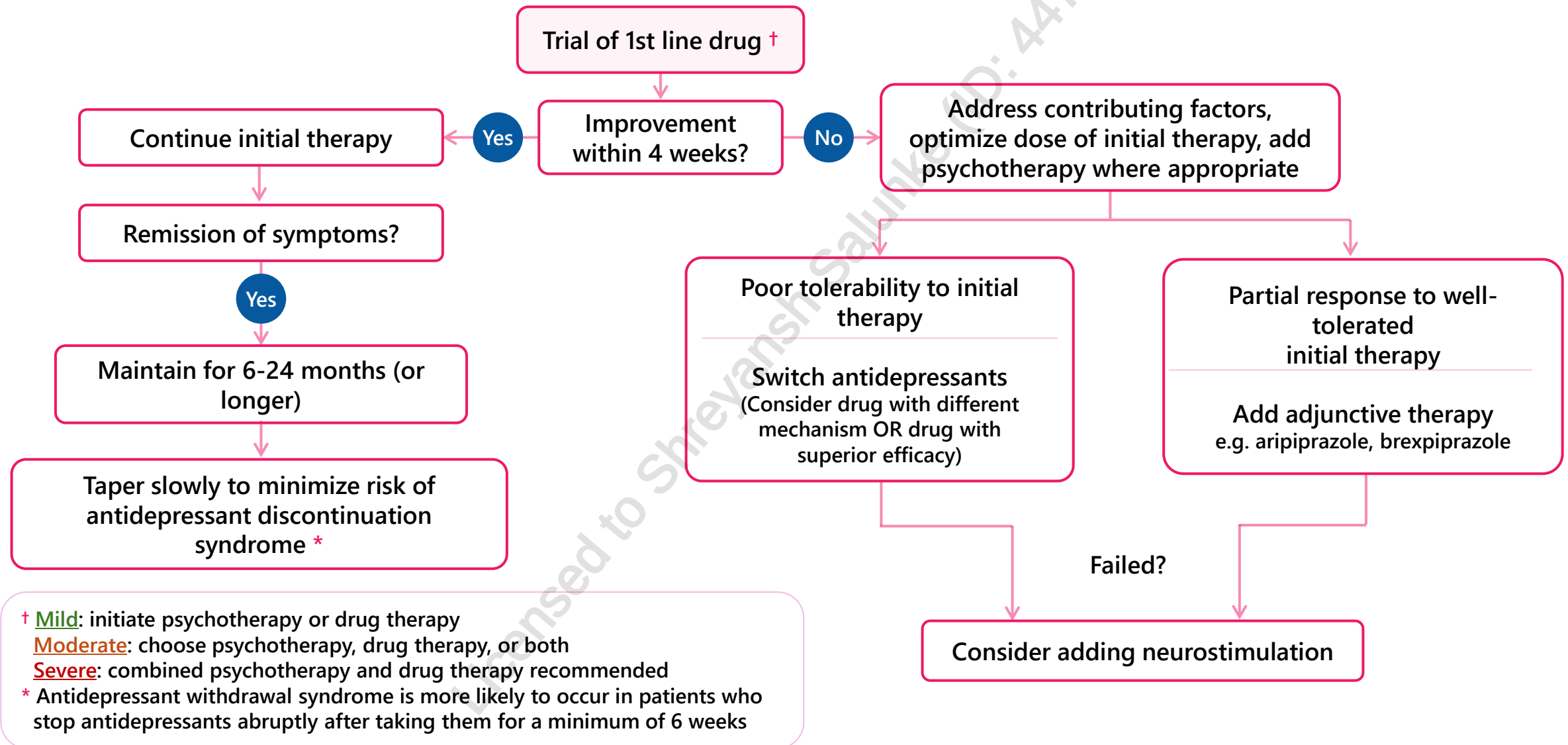
ECT (electroconvulsive therapy)

- Patients are placed under general anesthesia, electrodes are applied to their head, and a brief seizure is triggered (usually less than a minute)
- The mechanism is not fully understood, though there appear to be both cellular and chemical changes in the brain throughout a full course of ECT therapy
- First-line option in **severe and/or life-threatening situations** (suicidality, catatonia, psychosis)

Light Therapy

- Exposure to a light box with 10,000-lux intensity slanted toward the face for 30 minutes per day shortly after awakening
- First-line monotherapy for **MDD with a seasonal pattern (winter)**, improvement within 1-3 weeks
- Second-line option for **mild, non-seasonal depression**
 - group, etc.

PHARMACOLOGICAL TREATMENT ALGORITHM



PHARMACOLOGICAL TREATMENT

1st line

- Bupropion*
- Citalopram
- Desvenlafaxine
- Duloxetine
- Escitalopram*
- Fluoxetine
- Fluvoxamine
- Levomilnacipran
- Mirtazapine*
- Paroxetine*
- Sertraline*
- Venlafaxine*
- Vilazodone
- Vortioxetine*

*These drugs have superior efficacy

2nd line

- Moclobemide
- Quetiapine
- Trazodone
- Tricyclic antidepressants

3rd line

FYI

- Phenelzine
- Tranylcypromine

DURATION OF THERAPY

Acute Phase* (8-16 weeks until symptom remission):

- ✓ Address patient safety
- ✓ Treat to symptom remission and functional improvement
- ✓ Prevent suicide

**Overall: resolution of depressive symptoms*



Maintenance (6-24 months following acute phase) or longer if clinically indicated:

- ✓ Maintain symptomatic remission
- ✓ Restore functioning
- ✓ Prevent recurrence
- ✓ Discontinue treatment when clinically indicated

- All patients should be treated for 6-24 months after remission
- Treatment should be continued for ≥ 2 years if any of the following risk factors for recurrence are present:
 - Persistent residual symptoms (anhedonia, sleep problems, cognitive dysfunction)
 - History of childhood abuse
 - Increased severity of depressive episodes
 - Chronic depressive episodes
 - Psychiatric or nonpsychiatric comorbidities
 - Increased number of previous episodes
 - Poor social support
 - Persistent life events

PHARMACOLOGICAL AGENTS

SSRI and SNRI

Class/Drugs	Clinical Pearls
SSRI: • Escitalopram • Citalopram • Fluoxetine • Fluvoxamine • Paroxetine • Sertraline	• MOA: Selective Serotonin Reuptake inhibitors • Time to onset: 2-4 weeks • Start low and titrate up gradually • Fluoxetine is drug of choice in adolescents/children
SNRI: • Venlafaxine • Desvenlafaxine • Duloxetine • Levomilnacipran	• MOA: Serotonin-Norepinephrine Reuptake inhibitors • Time to onset: 2-4 weeks • Start low and titrate up gradually • Duloxetine has benefits for comorbid pain conditions • Desvenlafaxine has low incidence of sexual dysfunction

- Combinations with NSAIDs is associated with increased risk of gastrointestinal bleeding-consider addition of GI protective agent (i.e. PPI)
- Combinations with diuretics (e.g. hydrochlorothiazide) is associated with hyponatremia , especially in elderly)

PHARMACOLOGICAL AGENTS

NDRI and DAA

Class/Drugs	Clinical Pearls
NDRI: Bupropion	MOA: Norepinephrine and dopamine reuptake inhibitor Low incidence of sexual dysfunction - Reported to have pro-cognitive properties No weight gain - May increase anxiety Lowers seizure threshold Contraindicated in patients with history of a seizure disorder or recent history of anorexia or bulimia nervosa - Used with extreme caution where there is a history of head trauma - Caution with alcohol use, can also lower seizure threshold
DAA: Mirtazapine	MOA: Acts on both noradrenergic and serotonergic systems High rate of drowsiness, sedation and weight gain, increase in appetite and long-term metabolic risks Low incidence of sexual dysfunction

- Duloxetine and Bupropion are associated with drug-induced liver injury, which can occur up to 6 months after starting the medication

PHARMACOLOGICAL AGENTS

Serotonin Modulators

Class/Drugs	Clinical Pearls
Serotonin modulators: • Trazodone • Vortioxetine • Vilazodone	Trazodone • MOA: Potent postsynaptic serotonin receptor antagonist, weak serotonin reuptake inhibitory effects • Very sedating and rarely tolerated at therapeutic doses; used as sedative rather than antidepressant • Does not cause tolerance Vortioxetine • MOA: Combination of direct serotonin receptor modulation and serotonin transporter (SERT) inhibition • Reported to have pro-cognitive properties • Low incidence of sexual dysfunction Vilazodone • MOA: 5-HT reuptake inhibitor and partial 5-HT_{1A} agonist • Common side effects: mostly GI (nausea, diarrhea), headaches • Should be taken with food to ensure adequate absorption

PHARMACOLOGICAL AGENTS

TCA and MAOI

Class/Drugs	Clinical Pearls
TCA: • Amitriptyline, nortriptyline • Clomipramine, desipramine, imipramine • Doxepin	• MOA: Tricyclic antidepressants block the reuptake of serotonin and norepinephrine in presynaptic terminals • Time to onset: 2-4 weeks • Avoid TCAs in patients with history of overdose due to risk of cardiotoxicity, altered mental status and seizures
MAOI: • Phenelzine • Tranylcypromine • Moclobemide	• MOA: Monoamine Oxidase inhibitors • Associated with potentially fatal food interactions (hypertensive crisis possible when using tyramine rich foods like cured meats, mature cheeses, fermented products (miso, kimchi). Other foods containing tyramine are generally safe to consume • Caution when combined with medications affecting monoamine metabolism (e.g. dextromethorphan containing cough syrups, SSRIs, St. John's Wort)

SSRI SIDE EFFECTS

	ACH Effect (Dry Mouth)	GI Side Effects (Nausea)	Insomnia	Dizziness/ somnolence
Citalopram	++	++		++
Escitalopram	+	++	+	+
Fluoxetine	++	++	++	++
Fluvoxamine	++	+(constipation)	++	++
Paroxetine	++	++	++	++
Sertraline	++	++	++	++

0-9%: + 10-29%: ++ 30% or higher: +++

Sedation	Weight Gain	Sexual Dysfunction	QT ↑
More Favourable	Neutral	Less Favourable	Monitor QT
More Favourable	More Favourable	Less Favourable	
More Favourable	More Favourable	Less Favourable	
More Favourable	More Favourable	Less Favourable	
Neutral	Less Favourable	Less Favourable	
More Favourable	Neutral	Neutral	

More Favourable = preferred, to avoid the listed side effect

SNRI/NDRI/DAA/SEROTONIN MODULATORS SIDE EFFECTS

	ACH Effects (Dry mouth)	GI Side Effects (Nausea)	Insomnia	Dizziness/ somnolence	Sedation	Weight Gain	Sexual Dysfunction
Venlafaxine XR	++	+++	++	++	More Favourable	Neutral	Less Favourable
Desvenlafaxine	++	++	+	+	More Favourable	More Favourable	More Favourable
Duloxetine	++	++	++	+	More Favourable	Neutral	Less Favourable
Bupropion	++	++	++ (XL) ++/+++ (SR)	+(SR)	More Favourable	More Favourable	More Favourable
Mirtazapine	++			++	Less Favourable	Less Favourable	More Favourable
Vortioxetine	+	++	+	+	More Favourable	More Favourable	More Favourable

0-9%: + 10-29%: ++ 30% or higher: +++

More Favourable = preferred to avoid the listed side effect

TCA SIDE EFFECTS

	ACH Effect	Weight Gain	Sexual Dysfunction	GI Side Effects	Insomnia	Drowsiness	QT ↑	Orthostatic Hypotension
Amitriptyline	++++	++++	+++ / +++++	+		++++	+ / ++	+++
Clomipramine	++++	++++	++++	+	+	++++	+++	++
Desipramine*	+	+			+	++	+ / ++	++
Doxepin	+++	++++	+++			+++	+++	++
Imipramine	+++	++++	+++	+	+	+++	+++	++++
Nortriptyline*	++	+				++	+ / ++	+
Trimipramine	++++	++++			+	++++		+++

*Secondary amines are better tolerated than tertiary amines due to less side effects

slight: + low: ++ moderate: +++ high: +++++

OVERVIEW OF SIDE EFFECTS

Side effect profile	Drugs
Low sexual dysfunction	Bupropion, mirtazapine, vilazodone, vortioxetine, desvenlafaxine
Low weight gain	Bupropion, SNRIs, (less than SSRI – paroxetine has most weight gain)
Insomnia	SSRIs, SNRIs, bupropion, MAOIs, moclobemide
Sedation	Mirtazapine, quetiapine, TCAs, trazodone
Low seizure risk	Venlafaxine, trazodone, SSRI, moclobemide, MAOIs
High seizure risk	TCAs, bupropion

If adverse effects are severe, persist more than 2 weeks or are intolerable to the patient, consider:

- Lowering the dose or
- Switching agent

MANAGEMENT OF ANTIDEPRESSANT ADVERSE EFFECTS

ADVERSE EFFECT	MANAGEMENT
GI upset (Nausea, constipation, diarrhea)	Usually diminishes after 1-2 weeks of therapy • Nausea: use once daily dosing, take with food, have small frequent meals • Constipation: increase fiber intake, fluids, laxatives • Diarrhea: have small, frequent meals, limit caffeine, fluid replacement
Activation (anxiety, nervousness, agitation)	Usually diminishes after 1-2 weeks of therapy • Consider short term benzodiazepines only for extreme anxiety (avoid use whenever possible) • Severe agitation can be indicative of intolerance to therapy, switch to different drug class
Somnolence	Usually diminishes after 1-2 weeks of therapy • Consider bedtime dosing
Insomnia	Usually diminishes after 1-2 weeks of therapy • Consider daytime dosing • Counsel on sleep hygiene • Consider short term hypnotic (benzodiazepine or Z-drugs) only for extreme insomnia (avoid use whenever possible). Lemborexant can be used.
Weight gain	• Counsel on healthy diet and exercise routine, monitor weight and metabolic parameters • Consider switching to bupropion
Sexual dysfunction	Improvement in sexual function usually occurs in patients who achieve remission of depression • Consider switching to antidepressant least associated with sexual dysfunction • Consider starting PDE5 inhibitor in male patients
Serotonin syndrome	• Stop the drug and refer patient to hospital

SEROTONIN SYNDROME

- Adverse event occurring when there is **too much serotonin** in the body
- Typically occurs **when multiple serotonergic agents** are used **concomitantly**
- Substances to avoid when using other serotonergic agents:
 - Linezolid, MAOIs, the dye methylene blue
- Offending agents must be discontinued immediately, and patient should be referred to the emergency department

- **Irreversible MAOIs:**
Phenelzine and tranylcypromine. Highest risk of serotonin syndrome.
- **Reversible, selective MAOI:**
Moclobemide

"SHIVERS" is the mnemonic used to describe Serotonin syndrome symptoms

S	Shivering
H	Hyperreflexia
I	Increased temperature
V	Vital signs that are unstable (increased HR and RR, low BP)
E	Encephalopathy
R	Restlessness
S	Sweating

SWITCHING ANTIDEPRESSANTS

- **Direct switch:** stop the first agent abruptly and start the new antidepressant the next day
- **Taper and switch immediately:** gradually taper the first agent, then start the new agent immediately after discontinuation
- **Taper and switch after a washout:** gradually withdraw the first, then start the new after a washout period (washout depends on the half-life of the first drug)
- **Cross-taper:** taper the first (over 1-2 weeks or longer), and build up the dose of the new agent simultaneously

Examples of antidepressants

MAOI → other antidepressants	• 14-day washout period between discontinuation and initiation of new drug • Exception: Moclobemide (A washout period of at least 1 day should be observed before initiating another antidepressant (SwitchRx))
Other antidepressants → MAOI	• 14-day washout period • Exception: Fluoxetine washout period is 5 weeks
Fluoxetine → other antidepressants (excluding MAOI)	• Different protocols may be used in practice (e.g. cross-taper, stop/taper and switch after a washout) using an individualized approach and under close medical supervision

Consider patient specific factors such as severity of depression, sensitivity to adverse effects and anxiety

WITHDRAWAL/DISCONTINUATION SYNDROME

Rapid or abrupt discontinuation or dose reduction of most antidepressants may be associated with discontinuation syndrome, especially if taken for **6 weeks or more**

SNRI & SSRI: Withdrawal Symptoms

"FINISH"

- Flu-like symptoms
- Insomnia
- Nausea
- Imbalance
- Sensory Changes
- Hyperarousal

- Withdrawal usually lasts 3 days-3 weeks
- Can be prevented by counseling patients to not stop drug abruptly
- Patients should be counselled on withdrawal symptoms during drug tapering
- Resolve within 48h if SSRI/SNRI is reintroduced
- Always taper drugs to minimize side effects

Fluoxetine

- Lowest incidence of withdrawal symptoms due to long half-life

Bupropion

- Does not cause withdrawal

Paroxetine and Venlafaxine

- High risk of withdrawal due to short half-life
- Rapid onset of symptoms

PREGNANCY AND BREASTFEEDING

NEW
2025 GUIDELINES

- **Psychotherapy is 1st line**
- Pharmacotherapy in pregnancy is reserved for individuals who:
 - Have a severe symptoms
 - Those at high risk for relapse
 - Those who are unable to access or have an inadequate response to psychotherapy or
 - Those who prefer to use medication

PHARMACOTHERAPY IN PREGNANCY /BREASTFEEDING

First Line: citalopram, escitalopram, sertraline (best safety data)

- Memory Aid: **C**hildren **E**xist **S**afely

Second Line: bupropion, desvenlafaxine, venlafaxine, fluoxetine, fluvoxamine, duloxetine, mirtazapine

Third Line: TCAs, Trazodone, Quetiapine, Paroxetine (Avoid paroxetine in pregnancy - can cause cardiovascular malformations).

Avoid doxepin in breastfeeding → concern of infant sedation

POST-PARTUM DEPRESSION

- Approximately 1 in 5 Canadian women report symptoms of depression or anxiety in the postpartum period
- Screening is typically done with questionnaires (PHQ-9 or Edinburgh Postnatal Depression Scale), but diagnosis is not through questionnaire results alone
- Mild and moderate PPD may be treated with non-pharmacologic measures alone, but severe PPD typically will require a combination of pharmacotherapy and non-pharmacologic measures
- Consider if the patient is breastfeeding when selecting pharmacotherapy
- If there are severe symptoms or psychosis present, the patient should be referred for emergency care immediately

Zurzuva® (zuranolone) - FYI

- Approved by Health Canada in December 2025 specifically for moderate to severe postpartum depression – taken once daily for 14 days as a single course
- Contraindicated in pregnancy and does transfer into breastmilk (effect unknown)
- Mechanism is not fully understood, but is related to positive effects on GABA
- Causes somnolence; patients should not drive for 12 hours after taking a dose
- Common side effects: dizziness, drowsiness, sedation, headache
- Has a risk of dependence and misuse

COMPLIMENTARY AND ALTERNATIVE MEASURES

St. John's Wort

Mild depression – first-line (high level of evidence)

Moderate depression – second-line

Some effects on serotonin receptors and monoamine oxidase inhibition

Potent Inducer of CYP3A4 and P-glycoprotein; can decrease concentrations of many drugs

Acupuncture

Mild depression – second-line monotherapy

Moderate-severe depression – second-line adjunct

MONITORING PARAMETERS

Follow up within 1-2 weeks to assess safety, early improvement & tolerability

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Physical symptoms	Improvement in 2 weeks	3-4 weeks
Cognitive/emotional symptoms	Improvement in 4-6 weeks	Up to 1 year

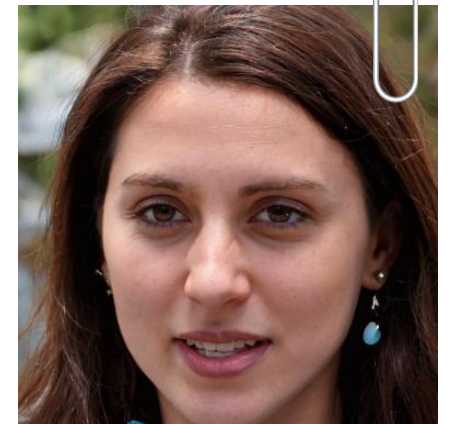
Safety Endpoints		
Parameter	Degree of Change	Timeframe
Side effects from medication therapy	Days-weeks	Ongoing
Suicidal thoughts and restlessness due to antidepressant initiation	Prevent (noticeable within 1-2 weeks of initiating therapy)	Ongoing (resolve once patient is stabilized on antidepressant)
Side effects from medication therapy	Days-weeks	Ongoing

TAKEAWAY

- Depression is a serious, but common, mental health condition characterized by a depressed mood that affects a patient's daily life
- Medications may take several weeks to take effect, and patients should be treated for 6 to 24 months, depending on severity and risk for recurrence of depression
- Treatment depends on the severity of symptoms (rated by self-administered questionnaire)
 - For mild depression, patients can trial psychotherapy first-line
 - For moderate depression, patients should trial either psychotherapy OR pharmacotherapy first-line
 - For severe depression, patients should be started on both psychotherapy AND pharmacotherapy
- First-line pharmacotherapy agents with superior efficacy: **bupropion, escitalopram, mirtazapine, paroxetine, sertraline, venlafaxine, vortioxetine**
 - Other first-line agents: citalopram, desvenlafaxine, duloxetine, fluoxetine, fluvoxamine, levomilnacipran, vilazodone
- Upon treatment discontinuation, patients should slowly taper off therapy (especially if on for 6 weeks) to minimize antidepressant withdrawal (FINISH)
- Patients should be educated on the signs, symptoms, and other medications that may cause serotonin syndrome, a life-threatening medical condition resulting from excessive serotonin (SHIVERS)
- In pregnancy and breastfeeding, psychotherapy and other non-pharmacologic measures are 1st line
 - If pharmacotherapy is needed, should utilize **citalopram, escitalopram, or sertraline**

CASE 1

MW, a 28-year-old female, has been feeling dramatically different from her usual self over several months and is presenting to an urgent care center for assessment. MW began experiencing symptoms of increasing fatigue and decreasing appetite over the last month. She has stopped hanging out with friends and she no longer enjoys the things she used to. For the past 2 weeks, she has been missing shifts at work as well as classes. She feels sad “all the time” and cries daily. Recently, she has started thinking of ending her life. MW works in retail, is a part-time student and is currently single. She quit smoking years ago and she drinks 3 glasses of red wine per day. She has a history of hypothyroidism. She takes levothyroxine 75 mcg po daily, Alesse® 1 tablet po daily, vitamin B12 1000 mcg po daily, and a multivitamin 1 tablet po daily. She is allergic to sulfamethoxazole/trimethoprim which presents as a rash. Today her labs show her TSH is 3.2 mIU/L (0.4-4.8), her BP is 129/73 mmHg, and her CBC is within normal limits.



CASE 1 QUESTIONS

1. Which of the following is LEAST likely to be a risk factor MW's symptoms?
 - a) Employment status
 - b) Female sex
 - c) Age
 - d) Alcohol use
2. Which of the following medications/supplements may be contributing to MW's symptoms?
 - a) Levothyroxine
 - b) Alesse®
 - c) Vitamin B12
 - d) Multivitamin
3. Which treatment strategy is LEAST appropriate for MW?
 - a) Cognitive behavioural therapy
 - b) Trazodone
 - c) Sertraline
 - d) Bupropion
4. Which drug would be LEAST appropriate to start?
 - a) Escitalopram
 - b) Duloxetine
 - c) Amitriptyline
 - d) Vortioxetine

CASE 2

Ms. MW returns to the health clinic 3 months later after being prescribed citalopram 20mg daily for major depressive disorder (MDD). She has been taking this dose for 6 weeks. She states she is feeling a bit better but is not back to normal yet.



CASE 2 QUESTIONS

1. Which next step would be the BEST recommendation for MW?
 - a) Optimize citalopram dose
 - b) Switch to quetiapine
 - c) Switch to venlafaxine
 - d) Add nortriptyline
2. What non-pharmacotherapy strategies should be recommended for MW?
 - a) Decrease alcohol intake
 - b) Physical exercise
 - c) Cognitive Behavioural Therapy (CBT)
 - d) All of the above
3. If MW were experiencing insomnia instead of feeling tired and sleepy, which medication may be most appropriate to use (instead of citalopram)?
 - a) Mirtazapine
 - b) Amitriptyline
 - c) Quetiapine
 - d) Paroxetine
4. If MW forgets to take citalopram for two days, what is she LEAST likely to experience on the third day?
 - a) Nausea and vomiting
 - b) Anxiety
 - c) Hallucinations
 - d) Flu like symptoms

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CHANGE LOG

April 2025

- Slide 21: Removed fluvoxamine and paroxetine
- Slide 30: New PPD slide
- Slide 38: Updated references

August 2025

- Minor formatting and phrasing changes throughout
- Slide 9: Added suicide safety plan information
- Slide 13: Added ECT information
- Slides 17-19: Split to break up information
- Slide 28: Added memory aid
- Slide 31: Added takeaway
- Slide 33: Removed 5th answer to align with MCQ format

November 2025

- Slide 3: Legend updated
- Slide 13: Expanded the definition of first-line use for ECT
- Slide 17: Addition of PPI
- Slide 18: Removed 'seizure' from the cautionary statement
- Slide 26: Added information on switching from fluoxetine to other antidepressants
- Slide 30: Updated follow-up duration

January 2026

- Slide 21: Added ++ signs to indicate that mirtazapine can cause drowsiness/somnolence

March 2026

- Slide 3: Added PPD
- Slide 19: Added vilazodone to serotonin modulators and moved TCAs and MAOIs to another slide (slide 20)
- Slides 21 and 22: Added note to clarify that 'more favourable' means drug is preferentially used to avoid the listed side effect
- Slide 24: Added vilazodone to low sexual dysfunction